

Two Simple Examples Use the sign test to compare two populations for significant differences for the paired data in Exercises 4–5. State the null and alternative hypotheses to be tested. Determine an appropriate rejection region with $\alpha \leq .10$. Calculate the observed value of the test statistic and present your conclusion.

4.

Population	Pair						
	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9
	+	+	+	-	+	+	+

5.

Population	Pair									
	1	2	3	4	5	6	7	8	9	10
1	6.9	8.2	8.9	6.3	2.6	3.5	7.5	6.0	7.6	1.2
2	7.1	8.5	5.1	11.4	8.4	9.1	10.8	6.9	8.3	10.9

$$\alpha = 0.1$$

$$H_0: P(A > B) = 0.5$$

$$H_a: P(A > B) \neq 0.5$$

$$\oplus = 6, \ominus = 1 \quad n = 7$$

$$P(X \geq 6) = P(X \leq 1) = P(X=0) + P(X=1)$$

$$= C_0^7 0.5^7 + C_1^7 0.5^6 \cdot 0.5 = 1 \times 0.5^7 + 7 \times 0.5^7 = 0.0625$$

$$2 \times 0.0625 = 0.125$$

$$p = 0.125 > \alpha = 0.1 \Rightarrow \text{non-reject } H_0$$

$$(A) H_0: P(A > B) = p = 0.5$$